

Development and application of deep learning-based method for riparian land cover classification using ortho-photos and LiDAR in flood simulation (the Asahi River area)

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Introduction

Background: Before this research, Okayama University hydraulic laboratory tried to use the Land Cover Classification (LCC) method based on data filtering. Due to the need of using multiple data filtering, this method is too cumbersome, and the results are quite different from the actual situation based on orthophoto, therefore this research tends to use the LCC method based on Deep Learning (DL) model with accuracy improvement.

Objective: This study is the first experimental evidence of using ALB and deep learning methods to extract hydraulic resistance parameters from orthophoto. And the water level results derived from the Asahi River in Okayama Prefecture, Japan 2018 flood simulation model using these parameters, has become almost 1-m closer to the field investigation rather than ALB-based method.

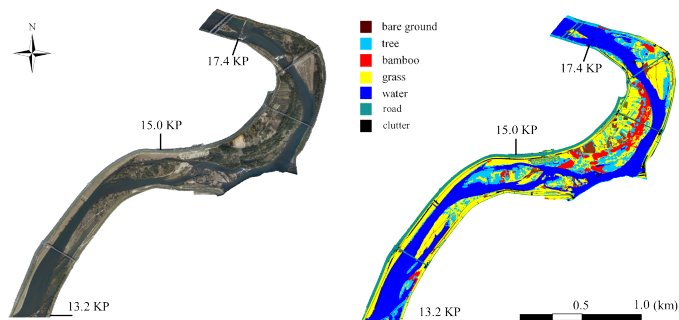


Fig. 1. Ortho-photo and True Label of the targeted area.

Methodology

- ◆Ortho-photo and True Label were used in the DL model (Fig.1).
- ◆LiDAR-assisted DL-based LCC method (Yoshida *et al.*, 2022).
- ◆Flood simulation using DL-based method-derived Hydraulic parameters
- ◆Comparisons of water level results derived from different LCC methods.

Key Results

- ◆DeepLabV3+ model has been modified for ALB data application to riparian LCC mapping (Fig.2).
- ◆Compared to the ALB-based method, the DL-based method is highlighted for distinguishing riparian vegetation species (Fig.3).
- ◆Hydraulic parameters derived using DL-based LCC results were reasonably used for flood simulation (Fig.4).
- ◆Our simulated water levels were much closer to field observations than those simulated using existing ALB-based methods (Fig.5).

Reference

Yoshida, K., Pan, S., Taniguchi, J., Nishiyama, S., Kojima, T., & Islam, M. T. (2022). Airborne LiDAR-assisted deep learning methodology for riparian land cover classification using aerial photographs and its application for flood modelling. *Journal of Hydroinformatics*, 24(1), 179-201.

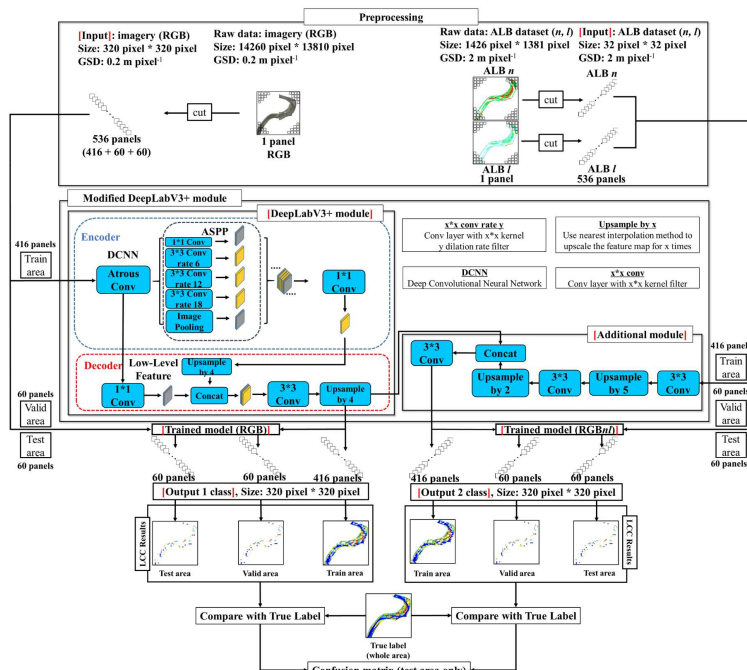


Fig. 2. Processing of mapping LCC with two DL methods.

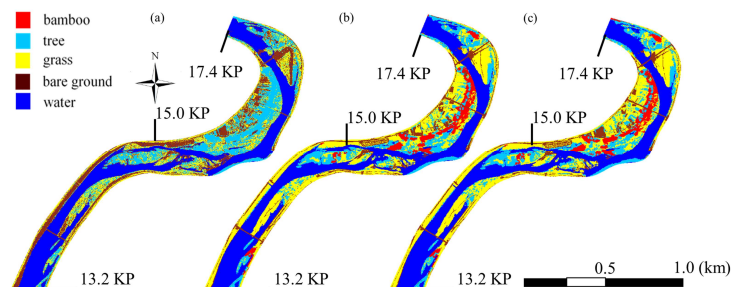


Fig. 3. Inferred Land Cover Classification results

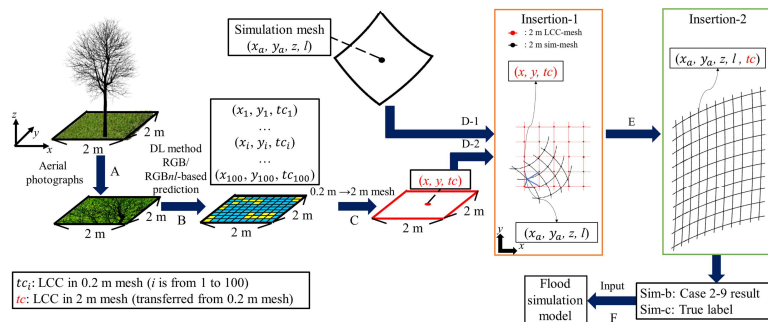


Fig. 4. Processing of transferring inferred LCC results for flood simulation.

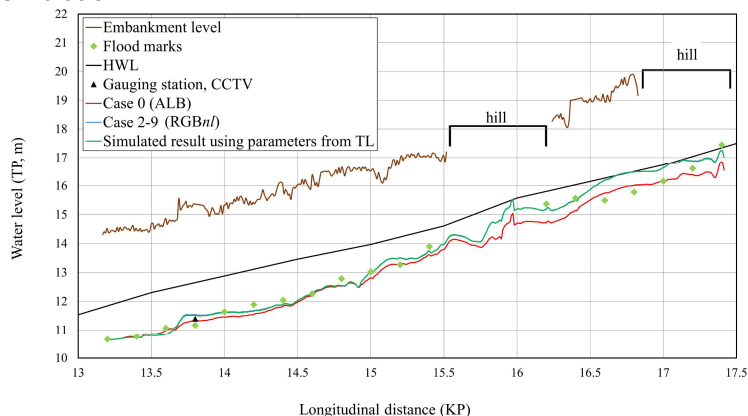


Fig. 5. Water level estimated from flood simulation results obtained using parameters of different LCC methods.